Measuring Early Word Exposure in Infants: A Brief Parent-Report Survey Captures Individual Language Input and Predicts Vocabulary Outcomes

Yuzhen Dong[1], **Charlotte Moore**[2,4], **Janani Ramadurai**[3,4], **Elika Bergelson**[1,4]

[1]Harvard University

[2]Vanier College

[3]University of North Carolina at Chapel Hill

[4]Duke University

Abstract

Individual differences in early language input are rampant but hard to quantify without resource-intensive naturalistic recordings, posing a challenge for testing links between individual experience and facets of language development like vocabulary growth. We tested whether a quick parent survey on infants' exposure to common nouns could reliably predict nouns' frequency in home-recordings (testing its validity) and children's own vocabulary (testing input-learning links). In Study 1 (n=44; 95% White, 5% mixed-race) we gathered monthly home recordings and exposure surveys every two months from 8-18mo. Parent reported exposure to each noun (16/timepoint, 5-point scale) was compared to the noun's relative frequency in infants' home-recordings, and to comprehension and production of each noun via parent vocabulary survey (CDI). Reported exposure significantly predicted noun frequency in home recordings. It also predicted both reported noun comprehension and production, even after accounting for age. Results from an age- and gender-matched cross-sectional sample (Study 2, n=264, 78% White, 11% mixed-race, 11% other) with the same vocabulary and exposure surveys were consistent, supporting generalizability. These findings suggest a brief word-exposure survey can reliably estimate individualized word input and knowledge, at least for highly common nouns. This method complements naturalistic recordings and corpus-based norms by capturing individual variation and enabling more scalable research across linguistic and sociocultural contexts.

Keywords

Language Environment; Word Learning; Parent Report; Vocabulary; Language Development; Naturalistic Recordings; Infancy

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Correspondence concerning this article should be addressed to Yuzhen Dong, 33 Kirkland Street, Cambridge, MA 02138, USA. yuzhendong@fas.harvard.edu.
Department of Psychology, Harvard University, Cambridge, MA, USA (YD, EB); Department of Psychology, Duke University, Durham, NC, USA (CM, JR, EB); Vanier College, Montreal, QC, Canada (CM). Department of Linguistics, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA (JR)

Introduction

Many factors are at play in early language development, including cognitive maturation, social interaction and quite critically, exposure to language. Research over the preceding decades has highlighted many links between aspects of children's language input, what they come to learn from it, and when (Anderson et al., 2021; Bergelson et al., 2023; Dominey & Dodane, 2004; Hoff & Naigles, 2002; Huttenlocher et al., 1991; Montag et al., 2018; Rowe, 2012; Rowe & Snow, 2020; Shneidman et al., 2013; Tamis-LeMonda et al., 2017; Weizman & Snow, 2001; Zammit & Schafer, 2011). But of course, children do not readily learn every word they hear. Researchers have long sought to understand which aspects of input make certain words easier to learn than others, and have identified relevant factors such as word length and concreteness (e.g., Belke et al., 2005; Coffey et al., 2024).

Unsurprisingly, word frequency, i.e., how often a word occurs, has emerged as a robust predictor of early word learning. Frequency effects permeate language learning, with links to not just lexical learning but phoneme recognition and syntactic construction as well (e.g., Ambridge et al., 2015; Huttenlocher et al., 2002; Teinonen et al., 2008). For early word learning in infancy (which is predominantly open-class words like nouns and verbs, Bergelson & Aslin, 2017; Hsu et al., 2017; Naigles & Hoff-Ginsberg, 1998; Waxman et al., 2013), word frequency within a given lexical class (e.g., within the category "noun") has been shown to be the most important predictor of its acquisition (Goodman et al., 2008; Naigles & Hoff-Ginsberg, 1998; Theakston et al., 2003; Theakston et al., 2004). The strong predictive effect of frequency remains even when controlling for other lexical features like concreteness (Ambridge et al., 2015; Swingley & Humphrey, 2018).

Given the central role of frequency in early language acquisition, researchers have developed various ways to measure word frequency in children's language environment. One widely used approach is to draw on large-scale corpora such as the Child Language Data Exchange System, or CHILDES (MacWhinney, 2000). CHILDES contains transcripts and media data (audio and video files) collected from conversations with children. The database is valuable for examining general trends in early language acquisition and has been used to establish frequency-based norms (e.g., Goodman et al., 2008). Researchers have created tools, for example, ChildFreq (Bååth, 2010), childes-db (Sanchez et al., 2019), and an associated R package (Braginsky et al., 2018), to enable easier access to the database and to retrieve frequency measures and a range of lexical properties. These tools typically rely on aggregated frequency estimates across multiple children, contexts, and age ranges. Implicit in their use is the assumption that such aggregated data provide a sufficiently representative approximation of the input experienced by a "typical" child in a given context or developmental window. Indeed, these generalized estimates have proven fruitful in accounting for a wide range of developmental phenomena, such as early noun learning, verb acquisition patterns, and the timing of vocabulary spurts (Ambridge et al., 2015; Goodman et al., 2008).

While aggregated frequency estimates may be well-suited for identifying broad trends or modeling population-level learning biases, they may be underpowered for explaining variation in individual developmental trajectories. Frequency counts aggregated across children do not reflect the distinct input profiles each child experiences, which might be highly variable especially in the early stages of life. For example, Swingley and Humphrey (2018) found that word frequency was a better predictor of vocabulary acquisition in true mother-child dyads than in randomly-generated simulated dyads. This finding suggests that input distributions for some word types are measurably idiosyncratic rather than extremely consistent across families. In this case, child-specific input measures are likely to offer greater explanatory power by tapping into individual variations in language input. In turn, this may lend insight into why or precisely how same-age children come to know different words.

That said, obtaining individualized frequency estimates can be challenging, and generally has required researchers to collect naturalistic recordings of speech from each child's environment, transcribe them, and then extract word counts from the transcripts. Notably, the CHILDES database consists of numerous individual naturalistic recordings collected by different studies (MacWhinney, 2000). Examining each recording at a child-level allows researchers to glean insights about individual differences in word exposure and other metrics about children's early language environments.

Obtaining individual-level frequency estimates in a longitudinal study can provide further insights into variability in early language development. One compelling example is the high-density case study by Roy et al. (2015), who reported on over 200,000 hours of audio and video recordings of a single child from birth to the age of three. Focusing on the period from 9 to 24 months, the researchers tracked the child's first production of nearly 700 words and examined predictors of word acquisition using a rich, multimodal corpus. Frequency of exposure emerged as a reliable predictor of earlier word production, along with other strong predictors such as contextual distinctiveness.

Naturalistic recordings have also provided insights into how different categories of words are learned. For instance, Meylan et al. (2013) used longitudinal corpora to show that children generalize over determiner usage early in development, suggesting that repeated exposure to structural patterns supports grammatical learning. Similarly, Liu et al. (2019) found that frequency and referential clarity both shaped early verb learning, and Nencheva et al. (2023) demonstrated that frequent and semantically rich caregiver input supports toddlers' acquisition of emotion words. Together, these studies highlight how frequency estimated from corpora serves as an important predictor of early word learning.

The examples provided above illustrate the value of naturalistic recordings to capture fine-grained, child-specific input data, and how they relate to children's subsequent production. However, despite their value, naturalistic recordings of children's home environments pose significant practical challenges, as data collection and transcription are highly resource-intensive. Transcribing a single hour of audio can take 6-12 hours of labor, and much longer if detailed annotations such as speaker tags or prosodic cues are added. These demands make it difficult to scale such methods across large samples or extended time periods. Moreover, most recordings, with the exceptions of Roy et al. (2015), capture only a small subset of a child's total language experience. Infants' exposure often occurs in clustered, context-specific episodes (Mendoza & Fausey, 2021), meaning that recording on one day (e.g., Wednesday) may miss recurring themes that occur on other days (e.g., Tuesday swim class). Access to some logistically-complex contexts to record in (such as daycare or other non-home settings) is also often limited, further restricting representativeness (Söderström et al., 2013). These constraints highlight the need for more scalable, flexible methods of estimating individual language input, particularly ones that can complement and extend beyond what is captured through naturalistic recordings alone.

One idea is to directly ask caregivers how often individual infants hears specific words. Parent-report questionnaires have been widely used in developmental research, most notably in assessing children's vocabulary size via the MacArthur-Bates Communicative Development Inventory (CDI, Fenson et al., 1994). The CDI is a standardized checklist in which caregivers indicate the words their child understands and/or produces, providing a widely validated measure of early lexical knowledge. This parent reported vocabulary knowledge has been shown to correspond highly with direct child assessment of these same skills (Ebert, 2017; Feldman et al., 2005; Marchman & Martine-Sussmann, 2002). More recently, using a longitudinal dataset, Egan-Dailey and Bergelson (2025) found that parent-reported productive vocabulary from the CDI at 1.5 years was the most robust predictor (among many) of a child's preschool language at 4.5 years of age.

The CDI has also been connected to children's language input across a range of studies. For instance, Goodman et al. (2008) linked CDI-derived age of acquisition data with corpus-based estimates of parental input frequency, showing that children tend to acquire more frequently heard words earlier, especially within lexical categories such as nouns. In the same vein, Bulgarelli and Bergelson (2024) recently reported that the acoustic variability with which children hear words produced helps predict when they themselves will say them. In the context of language exposure, parental reports have also been used to estimate the proportion of input languages in bilingual environments (e.g., DeAnda et al., 2016). However, it remains unclear how reliably parents can report on the frequency of exposure to specific words, or how often any particular word is actually heard by the child.

To our knowledge, no studies have validated parent-reported input frequency against more objective measures such as naturalistic recordings, which leaves open questions about the precision and utility of this kind of report. If parent-reported word exposure can predict both word frequency from naturalistic recordings and children's vocabulary as captured by the CDI, it could offer a scalable, individualized measure of early input. This measure could then bridge the gap between naturalistic recordings and large-scale frequency norms.

Building on this foundation, the present study introduces a brief parent-report word exposure survey, and evaluates whether parents' estimated input frequency converges with that derived from monthly home recordings from the same children. This tool was developed to provide quick, scalable estimates of infants' word exposure during the early months of vocabulary development. We focused our survey on concrete and imageable nouns, which are understood by 6-9 months (Bergelson & Swingley, 2012; Foushee & Srinivasan, 2024; Kartushina & Mayor, 2019; Parise & Csibra, 2012; Tincoff & Jusczyk, 1999) and dominate early vocabularies across many languages (Benedict, 1979; Bornstein et al., 2004; Fenson et al., 1994; Golinkoff & Hirsh-Pasek, 2008; Huttenlocher, 1974; Imai et al., 2008).

We administered the surveys within the Study of Environmental Effects on Developing Linguistic Skills (SEEDLingS) longitudinal project, which is a unique dataset combining naturalistic home recordings, parental surveys, and standardized vocabulary assessments for infants from 6 to 18 months. Using this rich dataset in Study 1, we asked whether a simple parental word exposure survey could predict the frequency with which words appear in infants' actual language environments (as captured by home recordings), and infants' vocabulary knowledge (as measured by parent vocabulary report from the CDI, Fenson et al., 1994). To further evaluate the generalizability of the measure, in Study 2 we also administered the survey and the CDI to an independent, age- and gender-matched cross-sectional control sample.

Collectively, this study addresses two overarching research questions. First, does parent-reported word frequency predict observed word frequency in naturalistic recordings of infants' language environment? And second, does reported word frequency predict an infant's vocabulary comprehension and production as measured by a second well-established parent report tool, the CDI?

Study 1: Testing the Word Exposure Survey in a Longitudinal Sample

Study 1 draws on the SEEDLingS longitudinal dataset, which includes naturalistic audio/video recordings, parent-reported CDI data, and our word exposure survey. This allows us to evaluate the extent to which parent-reported exposure aligns with objectively observed input, and whether it predicts children's vocabulary comprehension and production.

Study 1 Methods

The SEEDLingS dataset used here (as detailed in prior work e.g., Bergelson, 2017; Bergelson et al., 2019; Kalenkovich et al., 2024) was part of a broader study that investigated infants' early language environments, with a particular focus on nouns (which make up the bulk of the early vocabulary, Benedict, 1979; Bornstein et al., 2004; Fenson et al., 1994; Huttenlocher, 1974). The SEEDLingS dataset includes three main types of data (see Figure 1): (1) naturalistic audio and video recordings of infants' everyday interactions (collected monthly from 6-17 months); (2) eye-tracking data assessing infants' noun comprehension, which combined common nouns and hand-picked nouns from each child's preceding two months' recordings (collected every two months from 8-18 months); the eye tracking data is not analyzed in the current study but mentioned since it is connected to which items were queried by the exposure survey we focus on; (3) parent-report measures including demographic information, vocabulary assessments (collected monthly from 6-18 months), and the word exposure survey we focus on here (collected every two months from 8-18 months). Further details on the data collection procedure have been previously reported in prior work (e.g., Bergelson et al., 2019; Bergelson & Aslin, 2017; Kalenkovich et al., 2024; Moore & Bergelson, 2024). We provide critical aspects of the data collection methods here for readers' convenience as relevant for the current research questions and analyses. Some of the wording below is quite similar to the description of these methods in prior publications using this dataset, by necessity.

The present work focused on the naturalistic recordings and the parental surveys. We first describe the participants. We then detail our key data sources: the word exposure survey and three outcome measures that it aims to predict: (1) the recorded frequency with which speakers (excluding the target infant) said each target noun in naturalistic recordings; and (2) infants' comprehension; and (3) production of each word as reported on the CDI. Although all data were collected from the same group of children within the broader longitudinal framework, each data source was a distinct component, and is described separately.

Participants: Families were recruited from a mid-sized city in the Northeastern United States. Based on prior work, we targeted a sample size of 48, given components of the broader study that split the group into three experimental conditions of $n = 16$ each (the standard minimum sample size in the field). A total of 46 infants were initially enrolled, but two families withdrew soon after the onset of data collection. The final sample was thus 44 participants (21 females, including one pair of dizygotic twins). At the start of data collection, all infants were 6 months old (M = 6 months, 5 days; SD = 3.5 days). The last home visit occurred when the infants were 17 months, and the last in-lab studies occurred when the infants were 18 months.

All infants were exposed to English at least 75% of the time, were born full-term, and had no known hearing or vision impairments. The mothers in the sample were highly educated compared to national averages: 50% of mothers held graduate degrees, 41% had an associate's or bachelor's degree, 7% had some college, and the remaining 2% had at least a high school diploma. The sample was also racially homogeneous, with 95% of infants identified as white and 5% as mixed-race.

Informed consent was obtained from all parents at the start of the yearlong data collection period in a process approved by University IRB. Parents actively participated in various components of the study alongside their infants. Families received $340 and a small gift for their participation in the year-long study.

SEEDLingS Components

Word Exposure Survey. The word exposure survey used a 5-point ordinal scale (1 - Never, 2 - Less than once per week, 3 - A few times per week, 4 - Once a day, 5 - Several times a day) for parents to indicate how frequently their child heard each of the queried nouns. Each item was asked aloud, and parent ratings were noted by the researcher. We call this a *word* exposure survey, but note that we only queried nouns (which were tested during the eye tracking component of the broader study).

For each administration of the survey (every two months from month 8 to 18), parents were asked about 16 nouns, divided equally into two categories: common nouns and hand-picked nouns (see Figure 1). The 8 common nouns were pre-selected before the study for their high frequency in speech directed at North American English-learning infants; the same 8 common nouns were asked about across the sample for a given month, but a different set of 8 common nouns were queried at each timepoint. These common nouns all appeared on the CDI (e.g. "stroller", "diaper", and "ball".)

The 8 hand-picked nouns were selected from among the most frequent nouns identified in the infant's naturalistic audio and video recordings from the previous 2 months that were not already being tested as common nouns. This selection process factored in frequency, imageability, animacy, and other aspects relevant for the design of the eyetracking experiment in the broader study. To clarify, the 8 hand-picked nouns that were tested for an infant's 16-month visit would consist of 2 nouns each from their audio recordings at 14 months and 15 months, and 2 nouns each from their video recordings at 14 months and 15 months. Since hand-picked nouns were chosen based on individual language environments rather than general developmental norms, around half of them (62.50%) were not on the CDI. Nevertheless, they often included frequently-encountered nouns such as "boots" alongside somewhat less common ones like "tambourine." Over the course of a year of data collection, we collected exposure data for 41 distinct common nouns queried for all children (8 for each timepoint, with 7 repeats across timepoints) and 411 distinct hand-picked nouns (8 per timepoint per child, with 237 repeats across children). Further details about the handpicked items and the noun annotations more broadly can be found in the OSF preprint by Zhu et al. (2025), and in the SEEDLingS project Gitbook.

The word exposure survey was administered every two months when parents and infants visited the lab. Before each session, a research assistant prepared a response sheet listing the 16 nouns specific to the child's data collection session. This sheet was used to record parents' responses during the survey. Upon the family's arrival, the research assistant provided the parent with a laminated sheet displaying the 5-point ordinal scale, with each number accompanied by a clear description of its corresponding frequency category.

Parents were asked to rate how often their child heard each noun. In cases where parents hesitated between two response options, they were encouraged to select the option that felt closest, even if it was not perfectly precise. This approach ensured that all responses were recorded as whole integers. On average, it took approximately one minute to complete the word exposure survey.

CDI. The CDI is a widely used parent-report vocabulary checklist designed to assess early language development (Fenson et al., 1994; Frank et al., 2021; Mayor & Plunkett, 2011). It asks parents to indicate whether their child understands and/or says a broad range of common, early-learned words, providing estimates of both receptive (comprehension) and productive (spoken) vocabulary.

Participating families completed the CDI once per month from when their child was 6 months old until 18 months of age. To streamline the process and to reduce the reporting burden on parents, starting from the second administration, parents received pre-filled versions of the CDI with their responses from the previous month, digitally. This allowed parents to easily update any changes in their child's vocabulary each month. Parents were provided with the option to use hard copies or a digital administration; the vast majority opted for the latter.

We are examining CDI data from month 8 onward here only for several reasons. First, this provided consistency with the exposure survey and with the cross-sectional control sample, and second there are floor effects on CDIs at 6 and 7 months (which is part of the reason the instrument is normed for 8 months and older). From month 8 onward, 26 CDIs were not completed (from 12 infants), with more missing administrations in later months (18mo: 8 missing; 17mo: 6 missing).

Naturalistic Recordings. Participants contributed one full day of audio recording (up to 16 hours) and one hour-long video recording, collected on two separate days each month from when the children were 6 months until they were 17 months (see Figure 1). Collecting audio and video recordings on different days helped maximize the range of captured interactions within each family's natural environment. Due to technical issues, one audio file and one video file were missing from the final dataset, resulting in a total of 1,054 unique recordings (44 participants * 12 timepoints * 2 recording-types - 2).

To collect video recordings, researchers visited participants' homes to set up recording equipment. A camcorder (either a Panasonic HC-V100 or a Sony HDR-CX240) was placed on a tripod in the room where parents anticipated spending the most time during the following hour. If parents moved to another room, they were asked to bring the tripod with them to maintain continuous recording. Whenever possible, infants wore soft headbands or hats equipped with two lightweight Looxcie cameras (22 g each) to capture their point of view. Occasionally, infants refused to wear the headgear and parents wore the camera headbands instead. The cameras were positioned with one angled slightly upward and the other slightly downward to capture a broad visual field. After setting up the equipment, researchers left the home to minimize their influence on family dynamics. Each video session lasted approximately one hour, and the entire hour was analyzed (with concrete nouns annotated) for every month.

For audio data collection, parents were provided with LENA (Language ENvironment Analysis) recorders (LENA Foundation, Boulder, CO) to place in specially designed vests for infants. The LENA system is designed for daylong recordings of infants' language environments (Gilkerson et al., 2017). Parents were instructed to have their child wear the vest from wake-up to bedtime, excluding naps, bath times, or other situations where the vest was impractical. Parents retained the option to turn off the recorder at any time, though they were encouraged to capture as much of the day as possible. The resulting recordings captured up to (and modally) 16 hours of audio from a single day each month. For month 6 and 7, the full day was annotated for all concrete nouns as described below. We also algorithmically determined the top 5 "highest talk"[1] hours for each recording, and for later months annotated either the top 4 (months 8-13) or top 3 (months 14-17) of these. For standardization purposes described below, we analyze only data from the top 3 high-talk hours of each recording for analyses (i.e., using data from the same amount of time and the same subsampling process on all daylong audio recordings).

Trained research assistants listened to (and for video, watched) the recordings and annotated whenever a concrete noun was spoken (for details of this sampling and annotation procedure, see Kalenkovich et al., 2024, and the companion documentation). For each

[1]Adult and child talk were included in determining the highest talk regions (see Kalenkovich et al., 2024).

identified noun, the speaker identity (e.g., mother, brother), the context of the utterance (e.g., question, reading), and whether the object was present and attended to by the infant (yes or no) was coded. For the purposes of the current analyses, we focus solely on the frequency counts of each concrete noun across the recordings. Each noun token was coded in the form it was uttered (e.g., including plurals, diminutives, etc.). In order to group the related forms of the noun in each recording, each uttered noun was subsequently assigned a “basic level”, which generally corresponded to the most common version of the noun at the recording level. This is the word form that was tested in the eye tracking study, and appeared in the exposure survey. For example, if a family variably said “dog”, “doggy-woggy”, and “doggy” in a given recording, but said “doggy” the most (and dog, in its various forms, was one of the top nouns for that recording-type and timepoint), then “doggy” was the form of the word queried for the exposure survey.

Data Processing: Our analyses required several data processing decisions. First, all basic level nouns annotated from each recording were reconciled across the entire corpus by assigning a global basic level (more similar to the word’s lemma), in order to eliminate the family-specific variability in word forms (Moore & Bergelson, 2024). For example, an utterance that has a basic level of “doggy” or “dog” for different months of recordings would have a global basic level “dog” (see Kalenkovich et al., 2024 for details). We also applied the same process of global basic level conversion to nouns queried in the parental surveys, to allow for matching between home recording and the surveys.

Second, for parental responses “No”, “Understands”, or “Understands and produces” on the CDI, we created two binary variables: Comprehension (yes or no), and Production (yes or no). Additionally, certain words appear on the CDI multiple times due to distinct meanings (e.g., fish as animal and as food). Because our word exposure survey did not differentiate these senses, we collapsed CDI responses across meanings (for 7 unique nouns per CDI administration). I.e., a child was marked as understanding or producing a word if the response was yes for either sense of that word.

Third, regarding home recording frequency measures, we computed *relative* rather than *absolute* word frequencies to control for cross-infant variability in total noun input in the preceding two months of recordings. We computed individual level noun frequencies a specific infant heard rather than frequencies across all infants, since different nouns were queried for different infants. For each infant, the home recording frequency for each queried noun was calculated as a proportion of all nouns spoken in this infant’s previous two months of annotated audio (top 3 hours) and video (1 hour) recordings, excluding their own productions. Additional mentions of these nouns outside of the top 3 hours of audio recordings, or in any recording months beyond the previous two, were not taken into account (preliminary data exploration showed consistent results for various possible approaches here, see Appendix B; readers are welcome to investigate this within the shared data).

To give a concrete example, across the whole dataset, the most frequently heard noun across the two months preceding a study timepoint was “ball” for one 16-month-old infant participant. Ball was said a total of 199 times in this child’s home recordings across the 14 and 15 month timepoints. This same infant heard a total of 1091 noun tokens (across 169 noun types) in their 14 and 15 months recordings. Thus, the frequency for "ball" for this child was computed as 0.18 (199/1091).

Transparency and Openness: We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study, and we follow JARS (Appelbaum et al., 2018). Data were analyzed using R (Version 4.4.2; R Core Team, 2024) and the R-packages *blabr* (Version 0.25.1; Lab, 2025), *dplyr* (Version 1.1.4; Wickham et al., 2023), *forcats* (Version 1.0.0; Wickham, 2023a), *ggeffects* (Version 2.2.0; Lüdecke, 2018), *ggforce* (Version 0.4.2; Pedersen, 2024a), *ggh4x* (Version 0.3.0; van den Brand, 2024), *ggplot2* (Version 3.5.1; Wickham, 2016), *ggpubr* (Version 0.6.0; Kassambara, 2023), *glmmTMB* (Version 1.1.10; M. E. Brooks et al., 2017), *here* (Version 1.0.1; Müller, 2020), *lme4* (Version 1.1.36; Bates et al., 2015), *lmerTest* (Version 3.1.3; Kuznetsova et al., 2017), *lubridate* (Version 1.9.4; Grolemund & Wickham, 2011), *Matrix* (Version 1.7.1; Bates et al., 2024), *papaja* (Version 0.1.3; Aust & Barth, 2024), *patchwork* (Version 1.3.0; Pedersen, 2024b), *performance* (Version 0.13.0; Lüdecke et al., 2021), *purrr* (Version 1.0.4; Wickham & Henry, 2025), *readr* (Version 2.1.5; Wickham, Hester, et al., 2024), *readxl* (Version 1.4.4; Wickham & Bryan, 2025), *stringr* (Version 1.5.1; Wickham, 2023b), *tibble* (Version 3.2.1; Müller & Wickham, 2023), *tidyr* (Version 1.3.1; Wickham, Vaughan, et al., 2024), *tidyverse* (Version 2.0.0; Wickham et al., 2019) and *tinylabels* (Version 0.2.4; Barth, 2023).

All data have been made publicly available at the OSF through this link (https://osf.io/d2syb, Dong et al., 2025). This study was not pre-registered.

Study 1 Results

Analysis Plan: While centered around the research questions laid out above, the analysis we undertook was exploratory (and therefore not preregistered). We first examine the association between parents' reports of noun frequency based on the word exposure survey and the observed frequency of those nouns in the infants' naturalistic recordings from the preceding two months. This allowed us to assess the reliability of parental estimates in capturing actual language exposure, as sampled here. We employed mixed-effects modeling to address our first research question, assessing the extent to which parent-reported word exposure scores predict observed word frequency in infants home recordings. The word exposure survey response was mean centered to reduce the influence of multicollinearity. For the relative word frequency in naturalistic recordings, we employed a zero-inflated generalized linear mixed model, implemented in the R package glmmTMB (M. Brooks E. et al., 2017). This approach was chosen because the frequency distribution of words (including those in this corpus) is Zipfian (Zipf, 2016) and highly skewed. Relatedly, we also note that our frequency dependent variable is a proportion number between 0 and 1, and contained many zero frequencies (i.e. when a given noun was not said in some recordings).

We first added the word exposure score as a predictor of home recording frequency. We then added fixed effects of noun type (common vs. hand-picked) to test whether the relationship between home recordings' relative frequency and word exposure score was different for common vs. hand-picked nouns. Since the common nouns were common across all infants while the hand-picked nouns were individually selected for each timepoint, we expected that the latter may be more frequent. Random effects included random intercepts for words nested within subjects, and for age in months.

Next we explored the relationship between parent-reported word frequency (via the exposure survey rated on a 1-5 scale) and child vocabulary comprehension and production (binary outcome 1/0) as measured by the CDI. We used logistic mixed-effects models to address our second research questions (whether parent-reported word exposure scores predicted parent-reported vocabulary knowledge as measured by the CDI). In these models, we started with a fixed effect of infants' age, given its well-established role in vocabulary development independent of word exposure (Frank et al., 2021; Mayor & Plunkett, 2011). Then we sequentially added a fixed effect of word exposure score to test whether vocabulary knowledge could be predicted by word exposure score after controlling for age, and whether the relationship between vocabulary knowledge and word exposure might be different for different age groups. We also included by-subject random intercepts.

The final model was determined based on model comparison using the likelihood ratio test, as well as information criteria including *AIC* and *BIC*. When model comparison criteria conflicted, we adopted the simpler model, prioritizing parsimony and interpretability. The final models are presented below along with the model interpretation.

Word Exposure Survey: Before turning to the results based on each research questions, we first provide descriptive data about our key measure–the word exposure survey–in some detail.

There were a total of 41 distinct common nouns and 360 hand-picked nouns queried in the word exposure survey from month 8 to month 18, collapsing different versions of the nouns with our global basic level standardization process (described above; e.g., when collapsing the nouns "dog" or "doggie" hand-picked for different infants and/or different months). As noted above, since hand-picked nouns were selected based on their high frequency in each infant's past two months recordings, we expected hand-picked nouns to have higher exposure scores than common nouns.

Looking at the histogram in Figure 2, we first note that for both common and hand-picked nouns, parents used all five points on the exposure scale. Exposure scores are skewed toward the higher end of the scale. On our ordinal scale, a large proportion of queried nouns were reported to be heard by an infant several times a day (rating = 5), with only a few nouns marked as never heard by the infant. This skew was even more pronounced for hand-picked nouns, which were more frequently rated as 5. Common nouns received an average rating of 3.82 (SD = 1.19), while hand-picked nouns averaged 3.94 (SD = 1.14). There was a significant difference in exposure scores between common and hand-picked nouns (by Wilcoxon signed rank test). Aligning with our expectation, hand-picked nouns were rated significantly higher than common nouns, with a median difference of 0.12, $W = 16{,}910.50$, $p = .011$.

Word Exposure Score and Home Recordings Noun Frequency: As detailed above, frequency here was defined as a noun's relative frequency in a particular infant's naturalistic audio and video recordings from the past two months. This operationalization means that same noun would have different relative frequency for different infants, or for the same infant at different timepoints. Unsurprisingly, a histogram of the noun relative frequency shows a right-skewed Zipfian distribution (Zipf, 2016, see Figure 3).

Given the distribution of the ordinal word exposure survey scores and home recording frequency proportions[2], we calculated (non-parametric) Spearman's correlation between them. The correlation was statistically robust, but relatively modest (Spearman's rho = 0.27, $p < .001$, 95% CI [0.24, 0.3]). Given the skewed distribution of the exposure scores (and the skew in the opposite direction of the frequency distribution), we also ran this correlation by sampling an equal number of nouns at each exposure level (1-5) and performing a bootstrap correlation analysis with 1,000 resamples. This process resulted in a correlation of 0.34, $p < .001$, 95% CI [0.27, 0.4]. For further approaches to computing this correlation (including using adult frequency norms instead), which rendered similar results, see the Appendix B and C.

We next explored the nature of this correlation taking into account the two noun types (common and hand-picked), as a check of our noun selection process (see Figure 4). Descriptively, for both noun types, as exposure scores increased, the relative frequency of nouns increased. As expected, hand-picked nouns had higher frequencies than common nouns.

To analyze these interlocking relationships more formally, we ran a generalized linear mixed model. Using home recording frequency as the dependent variable, we started with a null model containing only the random effects: random intercepts for each noun nested within each subject (because each subject was asked about different nouns) as well as the infants' age in months. The model also accounts for excess zeros in the response variable, and uses the beta distribution to model the response variable. Model comparison revealed that adding word exposure score ($\chi^2 = 214.10$, $p < .001$) and the noun type ($\chi^2 = 160.60$, $p < .001$) significantly improved model fit, while adding in their interactions did not. The final model was therefore *relative frequency* ~ *exposure score* + *noun type* + (1|*subj*) + (1|*noun**) + (1|*month*)[3].

Fixed effects are shown in Table 1. The main effects of word exposure score (b = 0.15, $p < .001$) is significant. Higher exposure scores are associated with increased home recording noun frequency, as are hand-picked (as opposed to common) nouns (see Table 1 for details). For non-zero frequencies of a given noun type, a 1-unit increase in exposure score corresponds to 16.00% greater odds of an infant hearing the word in home recordings. Holding exposure score constant, hand-picked nouns have 121% greater odds of

[2]593 out of 4224 (14.00%) queried nouns in the exposure survey occurred zero times in the home recording data analyzed here. This is detailed further in Appendix A.

[3]Different random effects structures were needed for common and hand-picked nouns. Common nouns (queried for all infants for a given timepoint) required a fully crossed random effect structure. Hand-picked nouns (which differ for each subject) required a nested random effect structure (1 | subj/noun). To account for this, we created a new variable noun* that distinguished each potential overlapping hand-picked noun from each other (e.g. hand-picked noun "ball" for subject A was treated separately from the hand-picked noun "ball" for subject B in a different month). We then adopted a fully crossed random effect structure with this new variable.

appearing in home recordings compared to common nouns. See panel (b) of Figure 4 for a visualization of the predicted effects.

Word Exposure Score and CDI Vocabulary Scores: We next examined if parent-reported word exposure was associated with parent-reported vocabulary knowledge as measured by the CDI (comprehension and production). Given that we selected hand-picked nouns based on their high frequency for a particular infant, and links between input and vocabulary cited above, we expected that the hand-picked nouns would be more likely to be reported as comprehended and produced than common nouns.

We used logistic regression models to predict whether an infant was reported to understand and produce the noun of interest. Each model included parent-reported exposure survey scores for a given noun (1-5) and infant age (in months) as predictors. We also expected that the probability of comprehension and production would increase with increasing exposure scores and age. We are not looking at the effect of noun type in the logistic mixed effects models as it was not central to our research question and including it led to convergence failures. We retain it in the descriptives and figures for data transparency.

Word Exposure Score and CDI Comprehension. Starting with CDI comprehension, across the different months, parents were more likely to report that their infants understood a hand-picked noun (M = 0.67, SD = 0.47) than a common noun (M = 0.58, SD = 0.49), $t(248) = 6.42$, $p < .001$, 95% CI [0.06, 0.12]. This aligned with earlier findings that hand-picked nouns were more frequent, and with our expectation that greater frequency of exposure was associated with better understanding.

As the raw data and fitted logistic curve in panel (a) of Figure 5 show (omitting covariates and random effects), descriptively, across all ages, parent reported exposure score increased as the probability of CDI comprehension increased (seen in the smooth lines). We also find that across exposure scores, as the infant grew older, the probability of CDI comprehension increased.

To better quantify how age and word exposure scores were associated with CDI comprehension, we fit a logistic mixed effects model. We began with a model containing only the random intercepts for subject. Adding age ($\chi^2 = 1{,}013.93$, $p < .001$), word exposure score ($\chi^2 = 381.54$, $p < .001$), and an interaction term ($\chi^2 = 8.70$, $p = .003$) all improved model fit. The final model was therefore *CDI comprehension* ~ *age* * *exposure score* + (1|*subj*).

A summary of the fixed effects is available in Table 2. Holding exposure score constant, a one month increase of age was associated with a 0.49 increase in log odds, corresponding to 63.00% greater odds of comprehending a queried noun. Similarly, controlling for age, a unit increase in exposure survey score was associated with a 0.87 increase in log odds, corresponding to 139% greater odds of comprehension. Importantly, the two predictors interacted, suggesting that the effect of exposure score increased as the age of the infants increased[4]. The odds of comprehension associated with word exposure increase by 5.00% for each one-unit increase in age. See panel (b) of Figure 5 for predicted model results.

Word Exposure Score and CDI Production. We next looked at whether word exposure scores predicted CDI production. Because the onset of production is later than the onset of comprehension, we included in the production model only infants aged 12 months and older (those younger than 12 months were reported to produce on average <5 nouns on the CDI.)

Looking first at noun type, we found that as with comprehension, parents were more likely to report that their infants produced a hand-picked noun (M = 0.23, SD = 0.42) than a common noun (M = 0.15, SD = 0.36), $t(161) = 5.92$, $p < .001$, 95% CI [0.06, 0.11].

As visualized in panel (a) of Figure 6, the raw data and fitted logistic regression (not accounting for covariates) shows the relationship between the probability of CDI production and the exposure survey score at different ages. As infants grew older and as the exposure scores increased, the probability of word production increased (though, consistent with toddlers' lexical skill, production rates were far lower than comprehension rates). Quantifying these potentially interlocking factors, we next fit a logistic mixed effects model, beginning with a model containing only the random intercepts for subject. Adding the main effects of infants' age in months ($\chi^2 = 311.43$, $p < .001$) and word exposure score ($\chi^2 = 61.98$, $p < .001$) significantly improved model fit (but their interaction did not). The final model was thus *CDI production* ~ *age* + *exposure score* + (1|*subj*).

A summary of the fixed effects is available in Table 3. Holding exposure score constant, a one month increase of age is associated with 0.60 increase in log odds, corresponding to 81% greater odds of producing a queried noun. Similarly, for a given month of age, a unit increase in exposure survey score was associated with a 0.60 increase in log odds, corresponding to 82% greater odds of production[5].

Study 1 Discussion

In Study 1, we introduced a brief word exposure survey designed to provide a rapid estimate of children's language input, and we assessed how well it predicted (a) observed word frequency from our naturalistic home recordings and (b) infants' vocabulary knowledge (via parent-reported CDI). We find that the reported word frequency in the exposure survey correlated positively with the "ground truth" home recording word frequency, and that the reported word exposure predicted word comprehension and production as measured by CDI. The effect of word exposure on vocabulary was found alongside the expected effect of age for both word production and comprehension; for comprehension, word exposure and age also interacted significantly, such that the predictive effects of word exposure was stronger as the infant got older. We consider each of these results below. Overall, these findings suggest that the word exposure survey can serve as a quick yet informative tool for capturing meaningful input frequency data in research on early language acquisition.

Parent-reported Word Frequency Predicts Observed Word Frequency: Although parents made use of the full scale from 1 (never) to 5 (several times a day), responses were

[4]The main effects and interaction of age and exposure score in predicting CDI comprehension remained significant even after controlling for the relative noun frequency (see details in Appendix D).

[5]The main effects of age and exposure score in predicting CDI production remained significant even after controlling for the relative noun frequency (see details in the Appendix E).

not evenly distributed: most clustered at the more frequent end of the scale. This pattern was not particularly surprising given that we specifically tested high-frequency nouns, both common ones that most infants are likely to encounter frequently (e.g., “bird”, “spoon”) and hand-picked nouns that were selected due to their high frequency in the previous two months’ recordings (e.g., “drum”, “eggs”). Nevertheless, the finding that parent-reported word exposure closely aligns with home recordings word frequency suggests that parents have an intuitive awareness of their child’s linguistic environment: higher exposure scores predicted increased word frequency in the recordings. Moreover, word exposure scores remained a significant predictor after controlling for noun type (common vs. hand-picked). This is particularly noteworthy, since parents were not explicitly told which nouns were selected for them vs. common to all children when being given the survey.

The results support the use of parental reports as a reliable proxy for word frequency, especially when extensive naturalistic recording is not feasible. Given the considerable time and expense required for large-scale naturalistic data collection and annotation, the word exposure survey provides a practical alternative. Additionally, naturalistic recordings (especially shorter ones) typically capture only limited time periods and/or specific activities, which are unlikely to holistically represent a child’s language environment. Even in this relatively rich dataset, with both audio and video from different days, we largely relied on 8 hours of noun data (subsampled from ~34 collected hours of data) per child per timepoint (i.e., 8, 10 … 18mo.) as our “noun input” proxy for 2 months of language input. In Appendix B and C we consider different types of frequency estimate approaches, which all render quite similar effect sizes, but do suggest that parents likely use a more continuous sampling over their child’s estimated experience, drawing on exposure contexts not captured in the recordings.

The survey method offers the potential for broader coverage, capturing general language exposure without being constrained to specific contexts. That said, we do want to urge some caution in noting that our exposure survey only asked about nouns that are relatively high-frequency. While we were clearly able to capture some signal, it will be important in the future to also query parents’ estimated exposure to lower-frequency words and words from other grammatical classes. For such parts of the input, parent intuitions may be differentially reliable.

Parent-reported Word Frequency Predicts Vocabulary Comprehension and Production: We also found that reported word exposure predicted both comprehension and production scores on the CDI (alongside the expected effects of age). This suggests, consistent with prior research, that greater exposure to a word increases the likelihood that a child will understand and say it.

Finally, we also found evidence in support of the exposure survey’s sensitivity to how the queried nouns were selected. We expected that hand-picked nouns selected for a particular infant would be more frequent than common nouns for that same infant. Our results indeed found that both observed home recordings frequency and the parent reported word exposure scores of hand-picked nouns were higher than the corresponding scores for the common nouns. At the same time, however, interpreting the findings of the word exposure survey using this set of specially selected high frequency nouns still raises questions of generalizability. To address this limitation, it is essential to validate the effectiveness of the word exposure survey using a set of words that are independent of the tested infants' own language environment. One approach is to evaluate the survey's effectiveness in an independent control sample. In Study 2, we did, administering the same set of parental surveys to a separate group of cross-sectional infants matched to the original sample.

Study 2: Validating Word Exposure Survey in a Cross-Sectional Control Sample

Study 1 established that our word exposure survey was closely associated with and reliably predicted observed word frequency in infants' home environment, and their vocabulary knowledge. Study 2 tested the generalizability of these findings, taking advantage of the aspect of Study 1's design which separated common nouns (which are relatively high frequency for all children) and hand-picked nouns (which were picked for a particular infant given their high frequency in the preceding two months' home recordings).

To this end, we administered the same parental surveys (word exposure survey and CDI) to an independent control group that were age- and gender-matched to the SEEDLingS group, using the same set of nouns. This control group allowed us to assess whether the survey's predictive value for reported comprehension and production generalizes beyond the SEEDLingS cohort, and to compare performance across the two samples.

Study 2 Methods

Participants: 264 infants (128 females, 136 males) were recruited from a database in Durham, N.C., at Duke University. The mothers in the sample were highly educated compared to national averages: 53.80% of mothers held graduate degrees, 37.10% had an associate's or bachelor's degree, 3.80% had some college, 1.90% had a high school degree, and 0.80% had less then a high school degree, and the remaining 7 parents did not respond to this question. The sample was also largely white (78%), with 11% of infants identified as mixed-race, 5.70% as Black, 1.50% as Asian, 1.50% as other, and 6 parents did not respond.

Each control infant was matched to a single SEEDLingS infant + timepoint, based solely on their age and gender. For instance, if SEEDLingS participant 32 was a boy, then a boy control participant was found who was within 1 week of the right age at 8, 10, 12, 14, 16, or 18 months (i.e., 6 separate control participants, one for each timepoint, see Figure 7 for an illustration). All infants were born full term, had no known hearing or vision problems and were reported to hear English at least 75% of the time. Families received $5-10 depending on distance traveled to the lab, and a small gift for their lab visit for an eye tracking study (as in the SEEDLingS sample, the eyetracking data are not reported in the present analysis). Parents of 228 control infants completed the CDI, and parents of 257 control infants completed the exposure surveys. For analysis, we have 221 infants whose parents completed both surveys.

Design: The control group was tested cross-sectionally. Each control infant contributed data at a single timepoint, using the same exposure survey and CDI administered to the SEEDLingS group. The word exposure was always assessed the day they came in for the in-lab study, while for the CDI form (which is longer) parents were given the option to fill it at the visit or at home within about a week of the visit. We applied the same analysis procedures to both datasets to ensure consistency in how word exposure and vocabulary knowledge were assessed across the longitudinal and cross-sectional groups.

Study 2 Results

Analysis Plan: The results for Study 2 proceed similarly to the CDI portion of Study 1: we first summarize the word exposure survey results descriptively, then move to predicting CDI comprehension and production as a function of word exposure in this control sample. We employ the same visualization and model building approach described above. Finally, we compare results across Studies.

Since hand-picked nouns were not selected based on their high frequency for this control sample (by design), we expected that there would either not be any significant differences in the exposure ratings between common and hand-picked nouns, or that the hand-picked nouns would have lower exposure than the common nouns for this sample. Our prediction for the CDI was analogous: no difference relative to Study 1 for the common nouns, and either equivalent or lower word knowledge for the hand-picked nouns from Study 1. Finally, we predicted that the word exposure survey would predict CDI comprehension and production, just as it did in Study 1. As in Study 1, while we had predictions as laid out above, all analyses were exploratory and none were preregistered.

Word Exposure Survey: We collected exposure data for 41 distinct common nouns and 360 hand-picked nouns. As in Study 1, parents used all five points on the exposure scale. Common nouns received an average rating of 3.51 (SD = 1.34), while hand-picked nouns averaged 3.13 (SD = 1.45). There was a significant difference in exposure scores between common and hand-picked nouns (by Wilcoxon signed rank test). Aligning with our expectation, common nouns were rated significantly higher than hand-picked nouns, with a median difference of 0.38, $W = 5787$, $p < .001$. Figure 8 shows the distribution of the parental reported word exposure scores for common and hand-picked nouns.

Word Exposure Score and CDI Comprehension. Mirroring Study 1, we examined if parent-reported word exposure scores were associated with parent-reported vocabulary knowledge as measured by the CDI in the control infants. Consistent with our predictions, there was no significant difference (by t-test) between how likely parents were to report that their infant understood a hand-picked noun (M = 0.42, SD = 0.49) versus a common noun (M = 0.41, SD = 0.49), $t(220) = 1.01$, $p = .312$, 95% CI [−0.01, 0.04].

We next examined whether the word exposure survey predicted how likely a noun was to be reported as understood in this control sample. As the raw data and fitted logistic curve in panel (a) of Figure 9 show (omitting covariates and random effects) as infants grew older, and as exposure scores increased, the probability of CDI comprehension increased. To quantify this relationship, we fit a logistic mixed effects model.

We began with a model containing only the random intercepts for subjects. Adding age (χ^2 = 162.11, p < .001) and word exposure score (χ^2 = 485.01, p < .001) improved model fit, as did their interaction (χ^2 = 6.65, p = .010). The final model was thus *CDI comprehension* ~ *age* * *exposure score* + (1|*subj*), as in Study 1.

A summary of the fixed effects is available in Table 4. Controlling for exposure score, a one month increase of age was associated with 0.51 increase in log odds, corresponding to an odds ratio of 1.67. This means that each one-month increase in age was associated with 67% greater odds of comprehending a queried noun. Similarly, holding age constant, a unit increase in exposure survey score was associated with a 161% greater odds of comprehension. There was also a significant interaction between age and word exposure score, where the odds of comprehension associated with word exposure increase by 5.00% for each one-unit increase in age. The model predictions are shown in panel (b) of Figure 9.

Word Exposure Score and CDI Production. We next repeated our analysis for CDI production, including only infants aged 12 months and above, as in Study 1. In the control sample, at 12 months, infants were reported to produce an average of at least 9 words on the CDI.

Evaluated via t-test, there were no significant differences between how likely parents were to reported that their infant produced a hand-picked noun (M = 0.16, SD = 0.36) versus a common noun (M = 0.14, SD = 0.35), t(142) = 0.95, p = .344, 95% CI [−0.01, 0.04].

As for comprehension, panel (a) of Figure 10 shows the relationship between the probability of CDI production and the exposure survey score at different ages (depicted as raw data and fitted logistic curve without covariates). As infants grew older and as exposure score increased, the probability of CDI production increased.

To quantify these relationships, we fit a logistic mixed effects model. We began with a model containing only the random intercepts for subjects. Adding age (χ^2 = 38.58, p < .001) and word exposure score (χ^2 = 114.61, p < .001) improved model fit (but not their interations). The final model was therefore *CDI production* ~ *age* + *exposure score* + (1|*subj*), as in Study 1.

A summary of the fixed effects is available in Table 5. Holding exposure score constant, a one month increase of age was associated with 0.55 increase in log odds, corresponding to an odds ratio of 1.74. This means that each one-month increase in age was associated with 74% greater odds of producing a queried noun. Similarly, controlling for age, a unit increase in exposure survey score was associated with 140% greater odds of production. The model predictions are shown in panel (b) of Figure 10.

Cross Study Comparison: For our final set of analyses, we compared the results between SEEDLingS and controls across studies. As reflected in the tables and figures above, the results were quite similar across studies (with independent model selection processes arriving at the same model terms, and with relatively similar effect sizes). To examine this more concretely, since the SEEDLingS sample is longitudinal and the control sample is cross-sectional, we focused on a single month (18mo., i.e. the last timepoint) so that the combined sample is a consistent between-subjects comparison. We compared production at this age, as parental reported production was more robust compared to comprehension (Feldman et al., 2005; Frank et al., 2021; Houston-Price et al., 2007).

Since hand-picked nouns in the word exposure survey were selected based on their high frequency for the SEEDLingS group but not the control group, we expected that the parents of the SEEDLingS group would be more likely to report their infants producing a hand-picked noun than the controls, but similarly likely for common nouns.

CDI Production at 18 months: At 18 months, the reported probability of noun production in SEEDLingS (M = 0.41) was not different from control group (M = 0.36), $t(60) = 0.63$, $p = .533$, 95% CI [−0.12, 0.22]. The difference between noun types was not significant, $t(127) = 1.71$, $p = .090$, 95% CI [−0.02, 0.23], and is thus omitted in Figure 11.

To further quantify this relationship, we began with a model containing only the random intercepts for subject. Adding word exposure score ($\chi^2 = 35.67$, $p < .001$) significantly improved model fit, but not group (SEEDLingS vs. control) or noun type or any interactions among these terms. The final model is *CDI production* ~ *exposure score* + (1|*subj*).

A summary of the fixed effects is available in Table 6. A unit increase in exposure survey score was associated with 83% greater odds of production. Thus, at 18 months, there was not a significant difference between the two samples; see panel (b), Figure 11.

Study 2 Discussion

In Study 2, we assessed the effectiveness of the word exposure survey using an independent control sample, to test for generalizability.

Similar to the SEEDLingS cohort in Study 1, the word exposure survey results for common nouns showed a distribution clustered toward the higher end of the scale. This indicates that these nouns are generally high-frequency, as we would have expected from how they were initally selected (i.e. based on they frequency in aggregate data from relevant CHILDES corpora, as noted above). In contrast, responses to hand-picked nouns were more evenly distributed across the exposure scale among control parents, and had a lower mean exposure score. The response pattern is consistent with their selection process from other infants' recordings.

For vocabulary outcomes among the control infants, there was no significant difference in the average comprehension or production of common vs. hand-picked nouns. For CDI comprehension, we found significant main effects of both age and word exposure as well as their interactions across all control infants aged 8-18 months, same as SEEDLingS infants. Similarly, for CDI production among control infants aged 12mo. and beyond, we observed significant main effects of age and word exposure, mirroring the findings of Study 1. Thus the results of the cross-sectional control sample in Study 2 provides further evidence that the word exposure survey may be a useful predictor for assessing infants' word knowledge. Additionally, when combining data from both samples at 18 months, we again observed a significant effect of word exposure on production, further supporting the robustness of this relationship and the usefulness of the exposure survey.

Overall, the results of the cross-sectional control sample in Study 2 largely mirror that of the longitudinal SEEDLingS sample in Study 1. Taken together, this suggests that even in a sample for which we lack “ground truth” frequency information, the relationship between reported word exposure and both receptive and productive vocabulary appears robust. In turn, this underscores the potential utility of this survey for estimating individual children’s exposure to common nouns.

General Discussion

Broadly, the present study showed that a quick parent report word exposure survey was a robust predictor of how often infants heard frequent nouns in home recordings over a yearlong longitudinal study (Study 1). Moreover, we found that this survey predicted children’s early comprehension and production vocabulary in this same sample (Study 1) and in a second cross-sectional control sample of over 200 children (Study 2). We discuss the implications and limitations of these findings below.

Summary and Interpretations of Findings

In Study 1, we first found that the reported word frequency in the exposure survey correlated positively with the “ground truth” home recording word frequency. This correspondence with the more objective home recording word frequencies was an important aspect of evaluating the reliability of our word exposure survey. Given the importance of word frequency in lexical acquisition (Goodman et al., 2008; Swingley & Humphrey, 2018), the survey offers a promising, rapid, easy method that provides reliable estimates of linguistic input, at least in contexts like this one (a point which we return to below).

Moreover, unlike frequency norms derived from large-scale corpora, which can be challenging to generalize to very young children due to substantial individual differences in home language environments and acquisition trajectories (Swingley & Humphrey, 2018), our survey captures family-specific variability in language use. Researchers from different sociocultural contexts could potentially adapt this survey to test items that are more relevant to infants in broad-ranging circumstances, allowing them to quickly evaluate linguistic input without the extensive effort required by naturalistic recordings. Taken together, these results suggest that the word exposure survey can serve as a quick yet informative tool for capturing meaningful input frequency data in research on early language acquisition, at least for concrete, imageable nouns that dominate early vocabularies.

We also found in both Study 1 and 2 that the reported word exposure predicted word comprehension and production as measured by the CDI, in line with our predictions (though noting again that all analyses here are exploratory). The effect of word exposure on vocabulary was found alongside the expected effect of age for both word production and comprehension; for comprehension, word exposure and age also interacted significantly, such that the predictive effects of word exposure were stronger as infant age increased. The significant interactions suggest there may be sensitive periods during which linguistic input is particularly influential, highlighting the importance of future research into developmental timing effects within word learning (e.g., Bergelson, 2020).

Moreover, both Study 1 and 2 showed that CDI production scores were independently predicted by age and word exposure for infants aged between 12 and 18 months. Early word production (as well as comprehension) might also engage parents to provide additional language input, reinforcing vocabulary growth. Indeed, caregivers tend to speak more to infants who have started to produce words themselves (Dailey & Bergelson, 2023), which in turn, may then support further word learning.

Rethinking Measures: Parental Report and Frequency

While our results for exposure to vocabulary links were robust across both a longitudinal and cross sectional sample, they do both rely on parental report. The strong correspondence between these assessments may (at least partially) reflect parents' internal consistency or reliance on the same heuristic (e.g., how often they use a word themselves) when estimating exposure and vocabulary knowledge. This raises important questions regarding how parents treat these tasks.

Validity and Limitations of Parental Report Measures: The observed correspondence between parental ratings and objective frequency measures from naturalistic recordings suggests that, at the aggregate level, the survey provides a reasonable approximation of infants' exposure to noun input. But are the frequency estimates we are getting from the exposure survey really about the infants' experience? Or might parents be defaulting to just how common these words are in language in general? To test this, we looked at two adult English corpora[6], namely the Corpus of Contemporary American English (COCA, Davies, 2008), and SUBTLEX, (Brysbaert & New, 2009). In short, we found that word exposure ratings were moderately correlated with frequency norms from COCA (*rho* = 0.31) and SUBTLEX (*rho* = 0.29). The similarity of these correlations with those from our own early language corpus (*rho* = 0.27) suggests that parents might rely on broader lexical familiarity rather than child-specific experience alone when estimating word exposure.

Rather than this necessarily being an issue with the measure, it could just be that what parents are doing is reasonable, and that how common e.g., "potato" is in English is relatively consistent for families with young English learning children within a given relatively homogeneous socio-cultural context. Alternatively, it could suggest that the current framing of the survey, while designed to be more intuitive, makes it challenging for parents to be precise. The ordinal ratings also limit precise frequency comparisons across nouns or direct alignment with standardized frequency norms. Additionally, we note that we opted to treat our ordinal scales as rank-order, rather than absolute quantities. Given temporal imprecision in human memory, we presume caretakers are doing some sort of smoothing over their and their children's experience in estimating word exposure (as explored a bit in Appendix B and C). We leave to future research whether reframing

[6]See Appendix C for details.

the task may help align parent estimates more closely with child-specific language input, or whether finer-grained or continuous exposure scales would improve the precision of measurement while remaining usable for parents. That said, in our view, it's not obvious that 'absolute' versus 'relative' estimates of frequency over a given time horizon would render more interpretable results.

Moreover, we looked at the specific items on the CDI (1/0 on comprehension/production) that were on the exposure survey (1-5 rating). This allowed us to examine whether reported exposure to a word predicted reported understanding and/or production of a given word (and to characterize this in probability space using logistic regression). We found that the parent reported exposure scores showed convergence both with CDI comprehension (which is a somewhat less discerning measure, Frank et al., 2021; Houston-Price et al., 2007) and CDI production (which is more robust, Feldman et al., 2005). That said, our use case here aggregated individual-item level data, which is a relatively non-standard use (Fenson et al., 1994; Frank et al., 2021), though here we show that this does carry some signal (as others too have found, Hills et al., 2010, 2009; Nencheva et al., 2023).

Taken together, the reliance on parental reports in the present study (particularly for word comprehension) raises some concern about the extent to which the observed associations reflect true child knowledge versus reporting consistency. We would want more objective receptive knowledge measures to complement or validate the word comprehension findings in particular. In ongoing work, we plan to leverage the accompanying eyetracking data in this dataset to that end. Such approaches will help elucidate the extent to which parental estimates may be biased in their reflection of infant's word comprehension.

Reflections on How Frequency is Calculated: Our home recording frequencies were based on a narrow two-month window and computed as the proportion of target noun tokens among all noun tokens. This differs from more conventional frequency measures, which often include all word types. Our approach was a deliberate methodological choice, motivated by the dynamic nature of infants' language environments. As infants reach developmental milestones like crawling or grasping, their daily routines and conversational topics may change. Considering language input in a short window of two months may therefore allow us to capture potential bursts in word exposure. Moreover, this approach also aligns with how hand-picked nouns were selected for the word exposure survey, based on their high frequency in previous two months of home recordings.

Furthermore, only nouns were annotated in our dataset due to practical constraints and the focus of our study. As a result, calculating frequency as a proportion of all nouns within the same time window was a necessary compromise. We considered various frequency proxies, for example, calculating frequency across all available annotated audio hours rather than uniformly using top 3 audio hours for each time point. We found minimal differences using these approaches regarding the predictive ability of the word exposure survey[7].

[7]As detailed in Appendix B, in addition to the present approach, we also computed frequency based on all prior months, as well as all months of recordings. The predictive ability of the word exposure survey on vocabulary outcomes remained similar across these different frequency calculation methods.

Nonetheless, it is possible that our fine-grained, time-specific frequency measures, while more tailored to individual infants' experience, were too sparse to capture the patterns that parents were intuitively estimating. Future research could explore alternative aggregation approaches, such as pooling data over longer developmental windows or across infants of the same age, to better approximate the structure of parental intuitions about exposure. Moreover, our frequency measures did not factor in the different contexts and activities during which these nouns occur, and it is likely that specific activities (e.g., shared book reading or mealtime) could lead to bouts of different nouns. Understanding noun frequency in different contexts (and its ramifications on learning) could be an interesting topic for future work.

Limitations of Generalizability: It is important to note that our sample was relatively homogeneous in terms of language background (English speaking), maternal education (highly educated), and racial identity (mostly white). While this homogeneity reduced variability, it also limits the generalizability of our findings. Language exposure patterns, caregiver-report validity, and vocabulary development trajectories may differ in bilingual and multilingual families and across diverse socioeconomic and cultural contexts. For instance, different caretaking structures may influence caretakers' awareness of children's language exposure or vocabulary; we hesitate to guess how sizable such an effect would be, but welcome future work on this topic. That said, a recent large international study by Bergelson et al. (2023) found that the amount of adult talk and age were the most substantial predictors for child speech, while socioeconomic status and multilingualism did not explain how often children vocalized or how much adult talk they heard, though we note these measures are more coarse grained than those discussed here. In some contexts, combining parent-report tools with more direct observational methods may be especially important to accurately capture variability in early language input. Future work could fruitfully examine how the word exposure survey performs in more heterogeneous samples, and whether the observed relationships between reported exposure, home recordings, and vocabulary generalizes across different linguistic and socio-cultural environments.

Conclusion

Overall, our findings suggest that a brief word exposure survey can provide valuable insights into early language environments, making it a useful tool for language acquisition research. This 1-minute survey showed predictive value for both input frequency (as verified through home recordings) and vocabulary development (as measured by parental vocabulary checklist), across both longitudinal and cross-sectional samples. This finding sets the stage for future expansions across contexts, word types, and samples (e.g., more frequent or abstract words, Casey et al., 2023). By refining and extending this measure, we can improve our understanding of how early language exposure shapes vocabulary development in infants, potentially informing interventions and educational practices. At the same time, highlighting the value of the word exposure survey is by no means understating the importance of naturalistic recordings in capturing infants' language environment. When resources permit, naturalistic recordings still offer a much richer and more direct way to examine how words are used in context, allowing researchers to better capture the nuances of early language environments. Ultimately, these two approaches could be combined to offer complementary insights into infants' early language input across different linguistic and cultural contexts.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

This work was supported by NIH grant DP5-OD019812 to EB. The authors declare no conflicts of interest with regard to the funding source for this study.

We thank Lilliana Righter, Evgenii Kalenkovich, Hallie Garrison for their help with data collection and wrangling. We are also grateful to the undergraduate RAs and research staff at Rochester, Duke, and Harvard who helped collect, annotate, and clean the SEEDLingS corpus and its followups. Finally, thanks most of all to the families who made this research possible.

Appendices

The following sections contain the supplementary analyses associated with the main manuscript. There are five main sections:

- **A.** Explanation of zero frequency of queried nouns in the SEEDLingS corpus
- **B.** Correlations between word exposure scores and SEEDLingS frequency proxies
- **C.** Correlation between word exposure scores and adult frequency norms
- **D.** Word exposure predicting CDI Comprehension (controlling for relative frequency)
- **E.** Word exposure predicting CDI Production (controlling for relative frequency)

Appendix A Explanation of zero frequency of queried nouns in the SEEDLingS corpus

Across all queried nouns in the exposure survey over the different months (N=4224), there were 593 instances in which an infant's audio and video recordings from the past two months contained zero occurrences of the noun. Of these, 562 were common noun instances and 31 were hand-picked noun instances. These zero-frequency instances corresponded to 67 distinct queried nouns: 39 common and 28 hand-picked. The number of zero instances differs from number of distinct nouns with zero frequency, as the same noun can be queried multiple times across different months.

Of the 28 distinct hand-picked nouns that had a calculated frequency score of zero, 27 of them occurred outside of the top 3 hours of audio recordings transcripts used here to calculate frequency. That is, the hand-picked nouns were selected based on the full set of transcripts available at the time of selection (rather than just the top 3 hours for each audio used here). For months 6 and 7, this included the full videos and full-day audio recordings. For months 8 to 13, this included full videos and ~4 "high talk" hours from the audio recordings. For months 14 to 17, this included full videos and ~3 "high talk" hours from the audio recordings. The remaining zero frequency instance was a human coding error (the item selected, "vaccuum", did not actually occur in the given subject's recording that month).

Appendix B Correlations between word exposure scores and SEEDLingS frequency proxies

In our main analysis, we operationalized frequency as the relative frequency of a word in the past two months of SEEDLingS recordings. Here, we report correlations between exposure survey scores and other frequency proxies derived from the SEEDLingS corpus.

Focusing on frequency in the previous two months recordings (as in the main manuscript), we further considered available annotations from beyond the "top 3" audio hours, since this was the basis for the initial item selection, as described in the Appendix A. The frequency calculated from the full set of transcripts in the previous two months was moderately correlated with parent reported exposure survey scores, Spearman's *rho* = 0.29, $p < .001$, 95% CI [0.26, 0.32].

To explore whether parents may estimate word exposure based on a longer time frame, we calculated the correlation between the exposure survey scores and frequency calculated again from the top 3 hours of audio recordings and video recordings (as in the main manuscript) but this time from all preceding months and the current month (e.g., for month 12 we'd include recordings from months 6-12). The Spearman's correlation coefficient was 0.39, $p < .001$, 95% CI [0.36, 0.42].

Finally, we explored whether using the top 3 hours of audio recordings and video recordings (as in the main manuscript) from all timepoints (before and after those at which the hand-picked nouns were selected from) would change our results. The Spearman's correlation coefficient was 0.43, $p < .001$, 95% CI [0.4, 0.45].

Across all these analyses, while the exact magnitudes of the correlation coefficients varied (as expected, given changing amounts of data used to calculate frequency varied), all correlations were statistically robust, and around .3-.4 This suggests that parents' estimates of word exposure are robustly related to various proxies of word frequency derived from the SEEDLingS corpus. Thus our sampling choice (i.e., using data from the same amount of time (3 hours of audio and 1 hour of video) and the same subsampling process on all daylong audio recordings) seems unlikely to have had an outsized influence on our findings. That said, it is notable that the correlations were numerically strongest when considering longer time-horizons.

Appendix C Correlation between word exposure scores and adult frequency norms

In addition to frequency proxies computed directly from the SEEDLingS corpus, we explored whether parents may default to how common these words are in spoken American English in general. To this end, we calculated the correlation between the exposure survey scores and word frequency derived from two large-scale adult corpora: the Corpus of Contemporary American English (COCA) and the SUBTLEX corpus.

For word frequency derived from COCA corpus, the correlation coefficient was 0.31, $p <$ .001, 95% CI [0.28, 0.34]. For SUBTLEX frequency, the correlation coefficient was 0.30, p < .001, 95% CI [0.27, 0.32].

For both corpora, the correlations were statistically significant and similar in size to those from the SEEDLingS corpus, suggesting that parents might rely on broader lexical familiarity rather than child-specific experience alone when estimating word exposure.

Appendix D Word exposure predicting CDI Comprehension (controlling for relative frequency)

We began with a model containing only the random effects structure: random intercepts for word and for subject. Adding relative frequency (χ^2 = 112.92, p < .001), age in months (χ^2 = 1,080.93, p < .001), word exposure score (χ^2 = 251.31, p < .001) and an interaction term (χ^2 = 10.62, p = .001) all improved model fit. Further adding the noun type (common vs hand-picked) or any other interaction terms did not improve model fit. The final model is *CDI comprehension* ~ *relative frequency* + *age* * *exposure score* + (1|*subj*).

The model output was shown in Table D1. Importantly, the effects of age, exposure score, and their interaction remained significant even after the effect of relative frequency was controlled for.

Table D1

Fixed effects from the best-fitting linear mixed effects model predicting CDI Comprehension, controlling for relative frequency in the past two months SEEDLingS recordings. The intercept represents the odds of comprehending a word when the exposure score is at the centered mean of 3.88 at 12 months of age. Odds ratios are computed from the log odds.

term	log odds	odds ratio	SE	z	p.value
Intercept	2.12	8.29	0.37	5.79	<.001
Age	0.51	1.66	0.02	24.73	<.001
Exposure score	0.76	2.13	0.05	14.82	<.001
Relative frequency	0.34	1.40	0.05	7.11	<.001
Age x Exposure score	0.05	1.05	0.02	3.27	.001

Appendix E Word exposure predicting CDI Production (controlling for relative frequency)

We began with a model containing only the random effects structure: random intercepts for word and for subject. Adding relative frequency (χ^2 = 79.64, p < .001), age in month (χ^2 = 349.73, p < .001), and word exposure score (χ^2 = 24.61, p < .001) all improved model fit. Further adding the noun type (generic vs unique) or any interaction terms did not improve model fit. The final model is *CDI production ~ relative frequency + age + exposure score +* (1|*subj*).

The model output was shown in Table E1. Importantly, the effects of age, exposure score, and their interaction remained significant even after the very strong effect of relative frequency was controlled for.

Table E1

Fixed effects from the best-fitting linear mixed effects model predicting CDI Comprehension, controlling for relative frequency in the past two months SEEDLingS recordings. The intercept represents the odds of producing a word when the exposure score is at the centered mean of 3.88 at 12 months of age. Odds ratios are computed from the log odds.

term	log odds	odds ratio	SE	z	p.value
Intercept	−1.62	0.20	0.46	−3.50	<.001
Age	0.66	1.94	0.04	15.26	<.001
Exposure score	0.42	1.52	0.09	4.84	<.001
Relative frequency	0.65	1.91	0.08	8.52	<.001

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Public significance statement:

Understanding what language infants hear is important for supporting language development, but traditional methods are expensive and hard to scale. This study shows that brief parent-reports can reliably estimate infants' word exposure and predict vocabulary. Surveys like this could help researchers and practitioners query and support early language environments more widely.

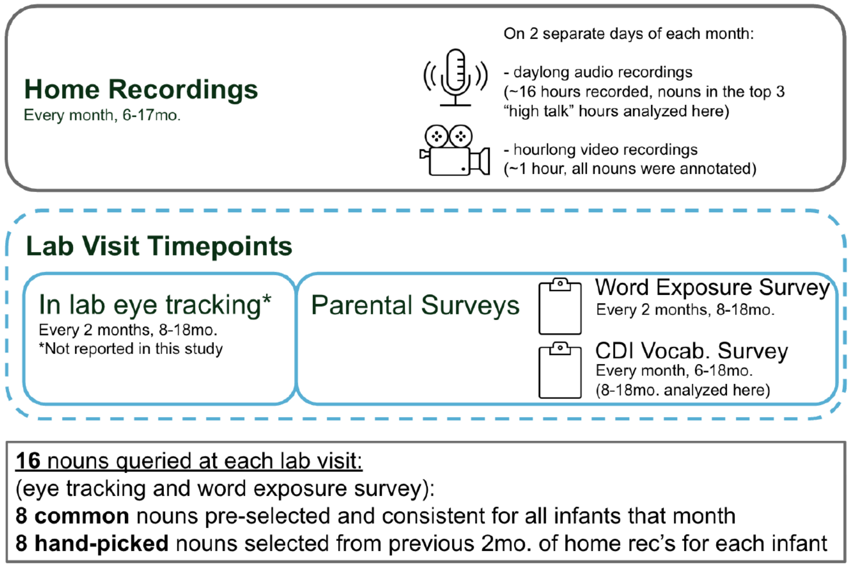

Figure 1.
Overview of the three components of the SEEDLingS dataset

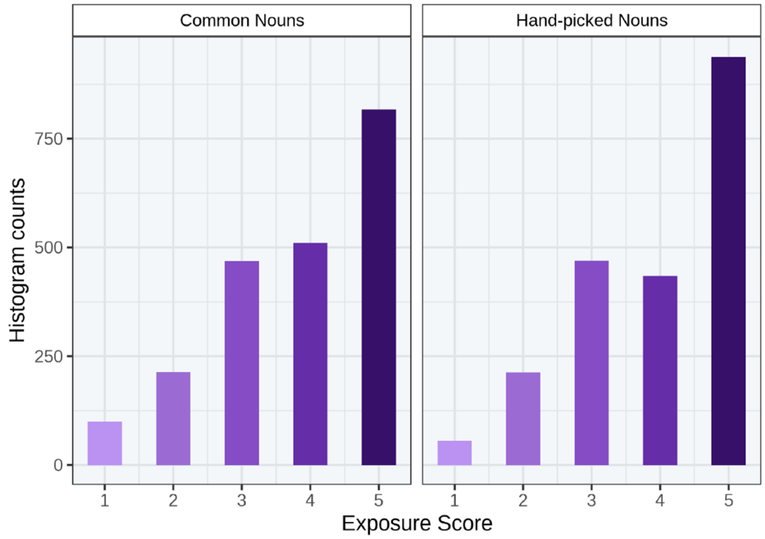

Figure 2.
Histogram of exposure survey scores for all queried nouns for the SEEDLingS group from 8 months onwards. No hand-picked words were queried at 6 months.

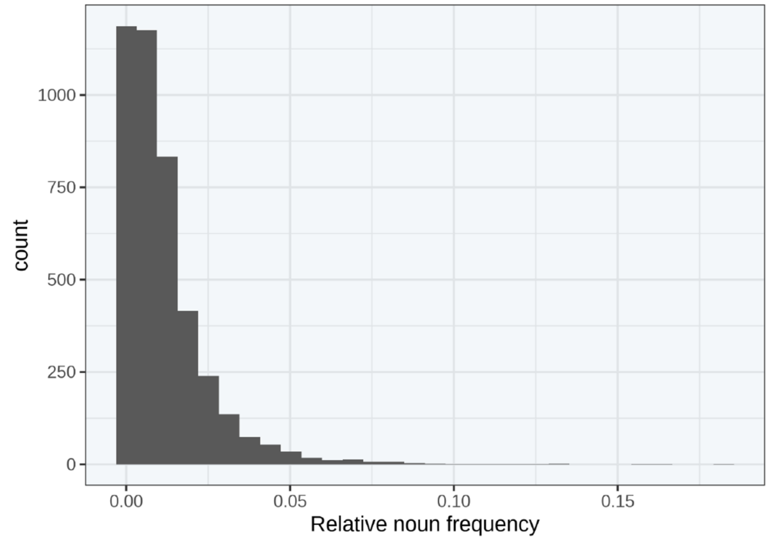

Figure 3.
Histogram of relative noun frequency from combined home recordings

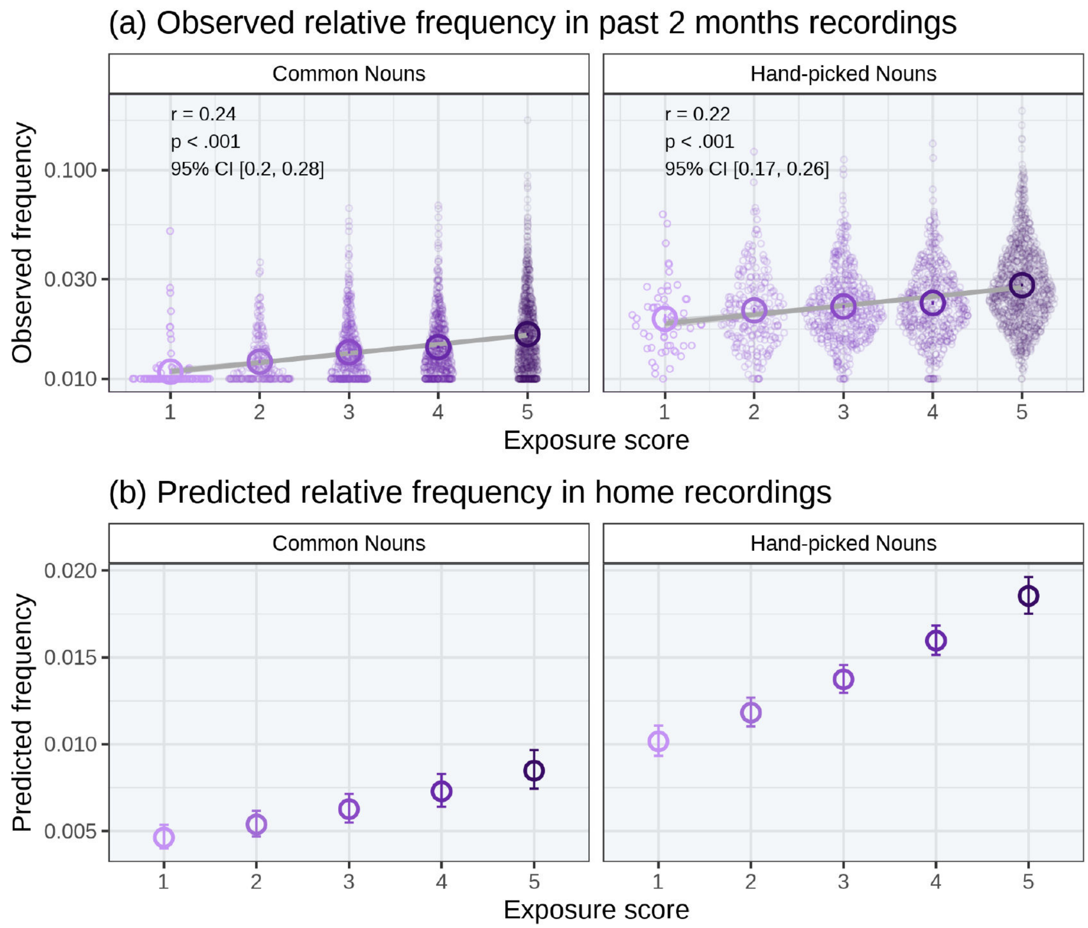

Figure 4.

(a) Observed and (b) predicted relative noun frequency (%) in past 2 months' home recordings as a function of exposure score and noun type. The relative frequency on the y-axis is represented on a log10 scale in (a) and on a cartesian scale in (b) for best visualization. The larger hollow circle represents the mean frequency at each exposure score. The gray area in (a) represents the 95% CI around the regression line. The correlation coefficients for common nouns and hand-picked nouns were slightly smaller than for all nouns combined. The error bars in (b) represent the 95% CI around the predicted values.

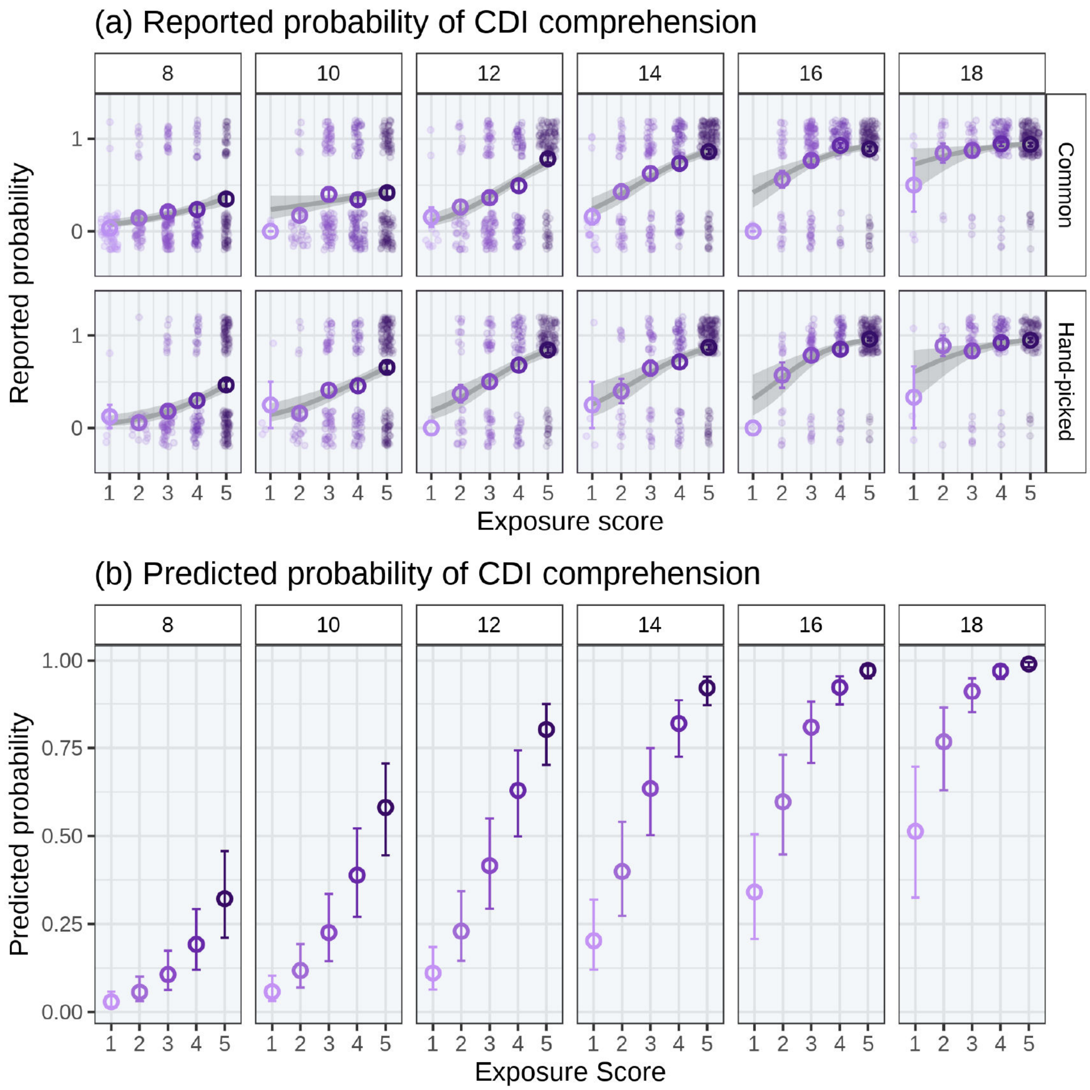

Figure 5.

(a) Monthly reported probability of CDI comprehension as a function of exposure score and noun type. Raw binary comprehension data (1 or 0) for each word for each infant at each month were jittered along the y-axis across exposure levels for visibility. The smooth line shows the predicted probability of comprehension, estimated using logistic regression. The gray area represents the 95% CI around the smooth regression line; (b) Monthly predicted probability of CDI comprehension as a function of exposure score based on the best-fitting model. The error bars represent the 95% CI around the predicted values.

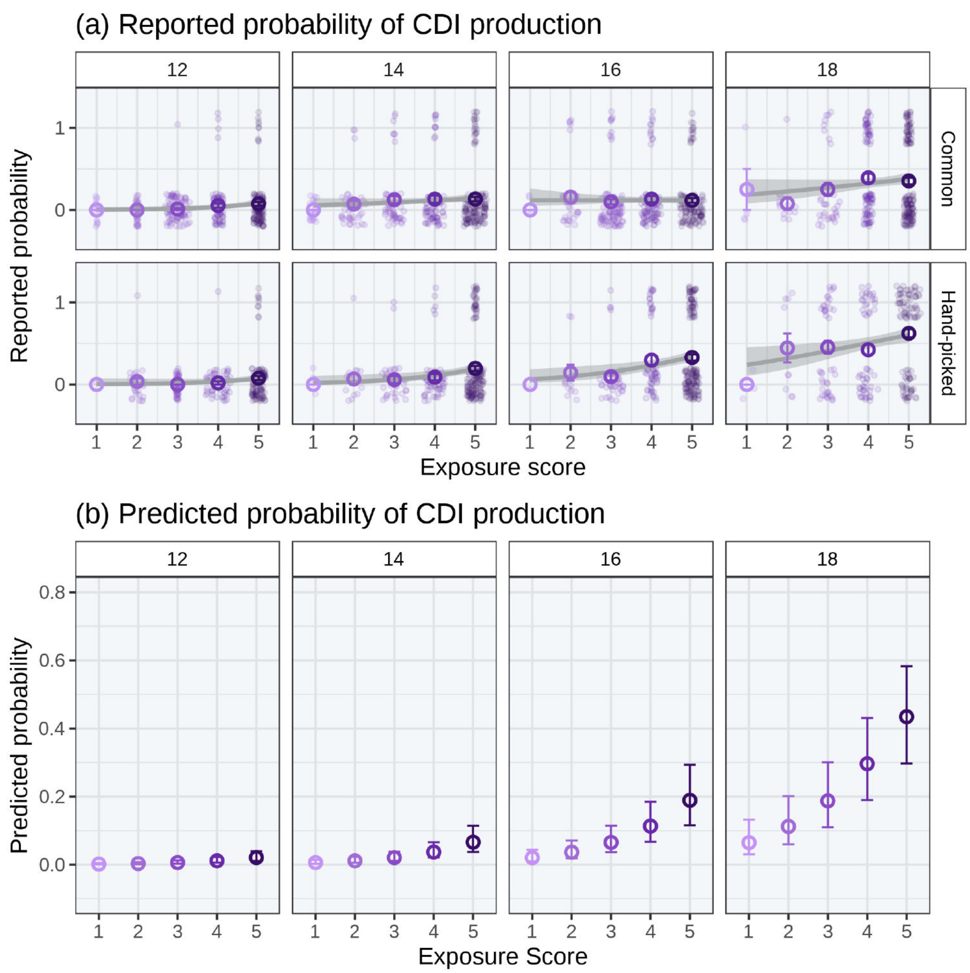

Figure 6.

(a) Monthly reported probability of CDI production as a function of exposure score and noun type. Raw binary production data (1 or 0) for each word for each infant at each month were jittered along the y-axis across exposure levels for visibility. The smooth line shows the predicted probability of production, estimated using logistic regression. The gray area represents the 95% CI around the smooth regression line; and (b) Monthly predicted probability of CDI production as a function of exposure score based on the best-fitting model. The error bars represent the 95% CI around the predicted values.

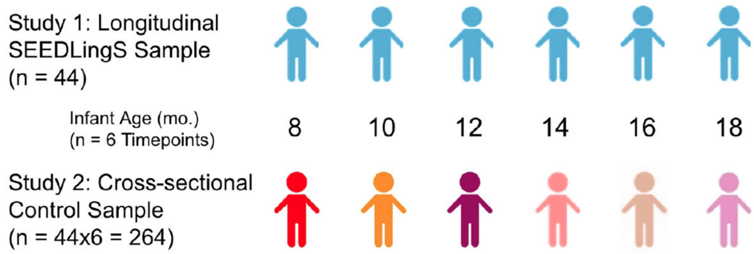

Figure 7.

Illustration of the control sample of Study 2 with reference to Study 1. For each timepoint (n = 6) for each longitudinal SEEDLingS infant in Study 1 (n = 44), an age- and gender-matched control infant was tested in Study 2 (n = 264). (The blue icon represents the same child across timepoints; the multi-colored icons each represent unique infants.)

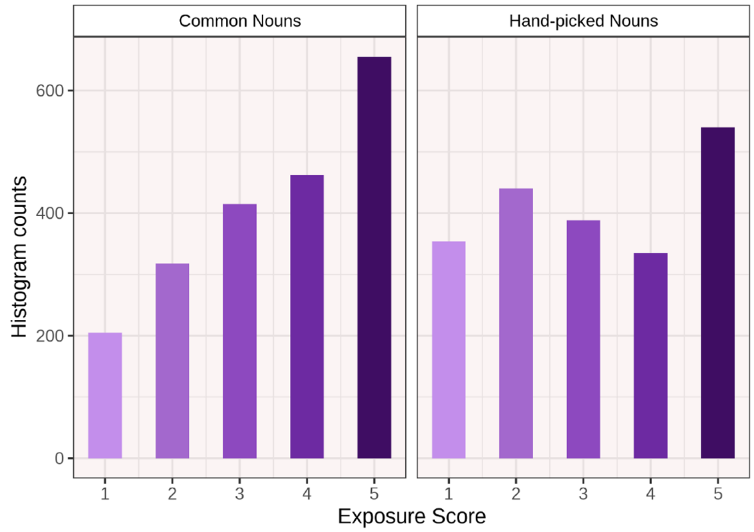

Figure 8.
Histogram of exposure survey scores for common and hand-picked queried nouns for the control group

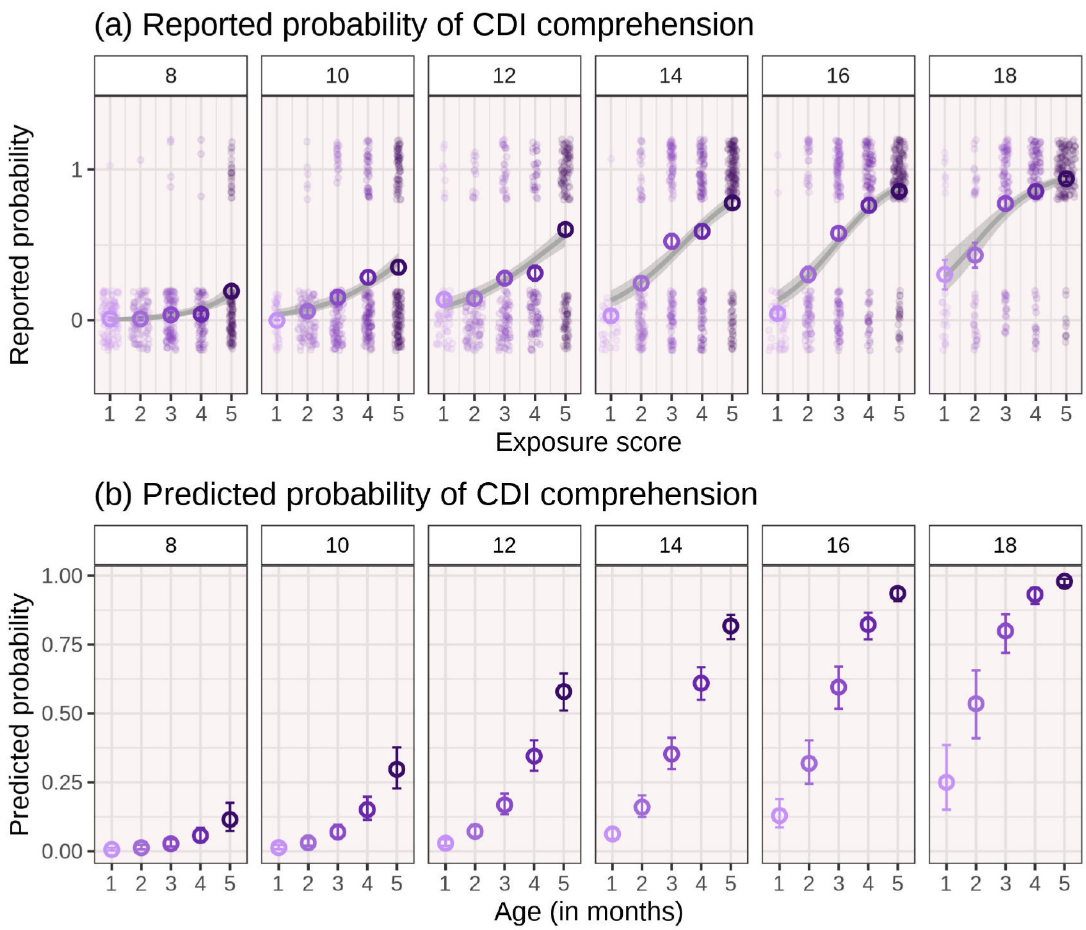

Figure 9.

(a) Monthly reported probability of CDI comprehension as a function of exposure score in the control infants. Raw binary comprehension data (1 or 0) for each word for each infant at each month were jittered along the y-axis across exposure levels for visibility. The smooth line shows the predicted probability of comprehension, estimated using logistic regression. The gray area represents the 95% CI around the smooth regression line; and (b) Monthly predicted probability of CDI comprehension as a function of exposure score based on the best-fitting model. The error bars represent the 95% CI around the predicted values.

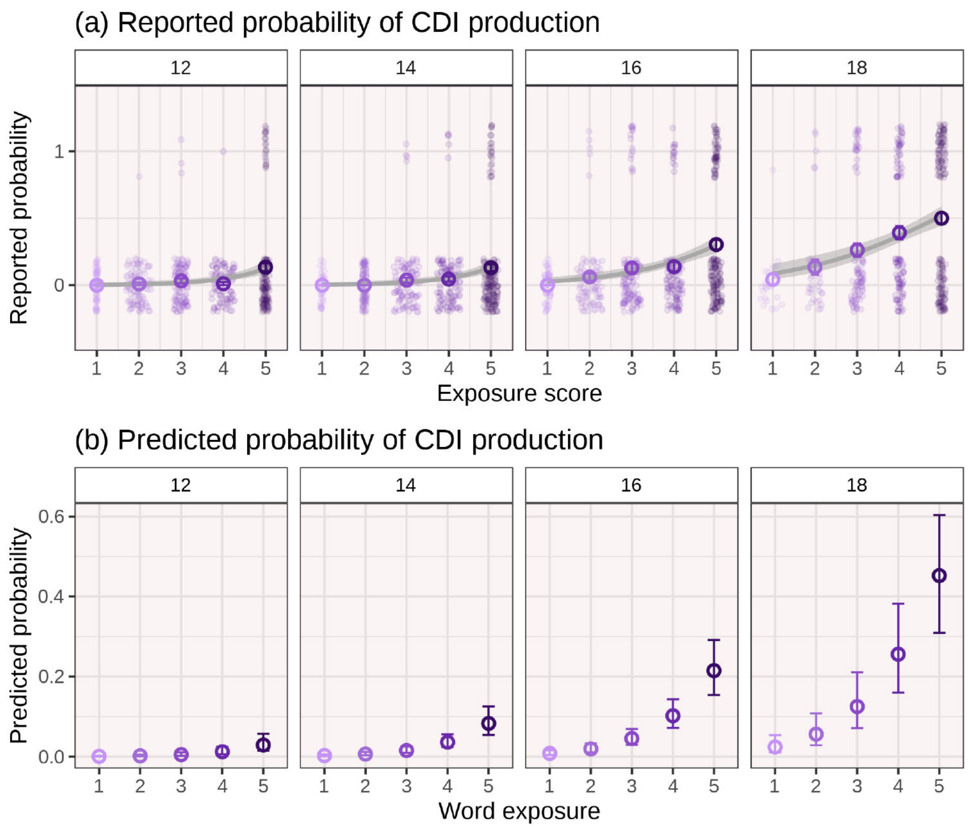

Figure 10.

(a) Monthly reported probability of CDI production as a function of exposure score in the control group. Raw binary production data (1 or 0) for each word for each infant at each month were jittered along the y-axis across exposure levels for visibility. The smooth line shows the predicted probability of production, estimated using logistic regression. The gray area represents the 95% CI around the smooth regression line; and (b) Monthly predicted probability of CDI production as a function of exposure score based on the best-fitting model. The error bars represent the 95% CI around the predicted values.

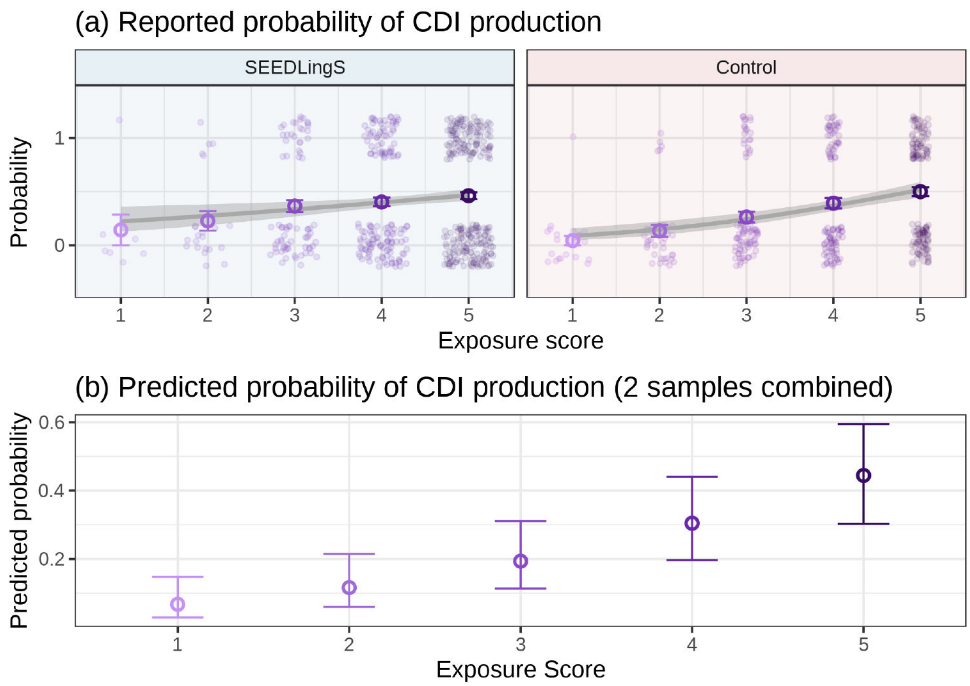

Figure 11.
(a) Reported CDI production as a function of exposure score at 18 month. Raw binary production data for each word for each infant at each month were jittered across exposure levels for visibility. The smooth line shows the probability of production, estimated using logistic regression. The grey area represents the 95% CI around the regression line; and (b) predicted probability of reported noun production as a function of exposure score for the SEEDLingS and Control sample. The error bars represent the 95% CI around the predicted values. The effect of the SEEDLingS/Control grouping variable is not significant and hence not included in the plot

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Table 1

Fixed effects from the best-fitting linear mixed effects model predicting relative nouns frequencies based on past two months' recordings. The intercept represents the relative frequency of a common noun when the exposure score is at the centered mean of 3.88.

term	log odds	odds ratio	SE	z	p.value
Intercept	−4.93	0.01	0.07	−74.33	< .001
Exposure score	0.15	1.16	0.01	13.35	< .001
Noun type	0.79	2.21	0.06	12.45	< .001

Table 2

Fixed effects from the best-fitting linear mixed effects model predicting CDI Comprehension. The intercept represents the odds of comprehending a noun when the exposure score is at the centered mean of 3.88 at 12 months of age. Odds ratios are computed from the log odds.

term	log odds	odds ratio	SE	z	p.value
Intercept	0.43	1.53	0.27	1.57	.118
Age	0.49	1.63	0.02	24.55	< .001
Exposure score	0.87	2.39	0.05	17.80	< .001
Age x Exposure score	0.05	1.05	0.02	2.96	.003

Table 3

Fixed effects from the best-fitting linear mixed effects model predicting CDI production. The intercept represents the odds of producing a noun when the exposure score is at the centered mean of 3.88 at 12 months of age. Odds ratios are computed from the log odds. p value is associated with the z statistic.

term	log odds	odds ratio	SE	z	p.value
Intercept	−4.51	0.01	0.34	−13.08	< .001
Age	0.60	1.81	0.04	14.72	< .001
Exposure score	0.60	1.82	0.08	7.37	< .001

Table 4

Fixed effects from the best-fitting linear mixed effects model predicting CDI comprehension in the control sample. The intercept represents the odds of comprehending a noun when the exposure score is at the centered mean of 3.49 at 12 months of age. Odds ratios are computed from the log odds.

term	log odds	odds ratio	SE	z	p.value
Intercept	−1.29	0.27	0.13	−9.65	< .001
Age	0.51	1.67	0.04	12.64	< .001
Exposure score	0.96	2.61	0.06	16.12	< .001
Age x Exposure score	0.05	1.05	0.02	2.59	.009

Table 5

Fixed effects from the best-fitting linear mixed effects model predicting CDI production. The intercept represents the odds of producing a noun when the exposure score is at the centered mean of 3.49 at 12 months of age. Odds ratios are computed from the log odds.

term	log odds	odds ratio	SE	z	p.value
Intercept	−4.98	0.01	0.41	−12.29	< .001
Age	0.55	1.74	0.09	6.07	< .001
Exposure score	0.88	2.40	0.10	9.16	< .001

Table 6

Fixed effects from the best-fitting linear mixed effects model predicting CDI production in SEEDLingS and Control samples combined. The intercept represents the odds of an infant producing a noun when the exposure score is at the centered mean of 3.99 at 18 months of age. Odds ratios are computed from the log odds.

term	log odds	odds ratio	SE	z	p.value
Intercept	−0.94	0.39	0.30	−3.12	.002
Exposure score	0.60	1.83	0.10	5.79	< .001